AI-Driven Stock Market Analysis and Portfolio Tracking Agent — Final Use Case (v2)

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0. How to Read This Document

This document has two kinds of requirements. **Owner Rules** are fixed business rules that the developer must not change. **Developer Decisions** are technical choices the developer makes and explains to the owner before building. Section 1 lists both clearly.

Important timing note: 5:45 AM IST is the **delivery deadline** of the report. It is not the time when the analysis starts. The analysis runs earlier in the background (see Section 8).

1. Owner Rules and Developer Decisions

1.1. Owner Rules (fixed)

1. The report must reach the configured email by 5:45 AM IST on every day the Indian stock market is open for trading.
2. Every original recommendation and forecast must be stored permanently and never overwritten or edited later.
3. Recommended stocks and stocks actually bought by the user must always be shown separately.
4. The system must never buy or sell shares by itself. Every transaction comes from user input or an explicitly authorized integration.
5. The system must never guarantee returns, invent missing data, or hide failures. If something fails, the report must say so.
6. A user's holding must stay visible until the user records a sale, even if the stock is removed from the recommendation list.

1.2. Developer Decisions (to be answered in a short plan before building)

The developer must send the owner a short written plan covering these questions:

1. Which main data provider and which backup provider will be used, and why?
2. How exactly will the five-day price forecast and the expected price range be calculated?
3. How will the system be tested on past data (backtesting), and what baseline will it be compared with?
4. Which database, scheduler, and hosting will be used?
5. How often will the background monitor run?
6. How will the user log in to the dashboard safely?
7. How will the system detect and fix wrong or missing data?

2. Use Case Overview

Use Case Name: AI-Driven Stock Market Analysis, Prediction, and Portfolio Tracking Agent

Objective: Build an automated AI agent that analyzes the Indian stock market, finds stocks with possible price growth over the next five trading days, predicts a likely price range, and keeps watching all previous recommendations. The system also has a Portfolio Dashboard where the user records real purchases and sales. The system tracks those holdings every day, compares real results with the original forecasts, and emails a detailed HTML and PDF report every market morning by 5:45 AM IST.

Primary Goals: The system must analyze the Indian market daily, find stocks with possible growth over five trading days, create new recommendations every trading day, keep every original forecast safe, watch earlier recommendations and warn about negative changes, check completed forecasts against real outcomes, let the user record real investments, track portfolio value and profit or loss, and send reports automatically. It must work as a monitoring and tracking system, not just a daily list of stocks.

Regulatory Note: In India, giving paid or public investment advice may fall under SEBI rules for investment advisers and research analysts. If the tool is only for the owner's personal use or learning, this is normally not an issue. If it will be shared with others or sold, the owner must check SEBI requirements with a qualified professional first.

3. Scope and Configuration

Parameter	Requirement
Target Market	Indian stock market
Exchanges	NSE and BSE
Stock Universe	Nifty 50 and Nifty 100 constituents (each stock analyzed only once per exchange identifier)
Prediction Horizon	Next 5 trading days
Background Monitoring	Continuous (light checks every few minutes, frequency chosen by developer)
Main Analysis	Once daily after market close, plus a final refresh before the report
Report Delivery	By 5:45 AM IST, only on days when NSE is open for trading
Timezone	Asia/Kolkata
Report Formats	HTML and PDF
Delivery Method	Automated email
Investment Input	Dedicated Portfolio Dashboard
Storage	Persistent history of recommendations, forecasts, transactions, and snapshots
Cost Basis Method	FIFO (first in, first out)
Execution Mode	Automated. User input is needed only for investment transactions and setup

The system should allow adding more indices or markets later without a full redesign. Holdings that are outside the Nifty 100 must still be tracked and valued if the user records them.

4. Market Data Collection

4.1. Market Data

The system must collect daily open, close, high, low, volume, latest available price, historical returns, historical volatility, corporate actions (splits, bonus shares, dividends), index prices, sector performance, the exchange trading calendar and holidays, and adjusted historical prices where suitable.

For every important analysis, the system must store the **timestamp and source** of the data used.

4.2. Data Source Rules

The developer chooses the providers. Official NSE and BSE data, Yahoo Finance, and Alpha Vantage are options, but the developer must check coverage, limits, licensing, and cost, and must not assume any provider covers everything.

The system must use **one main source and one backup source**. It must run a data-quality check comparing them. If closing prices differ by more than a set percentage, or data is missing, delayed, or unusual, the stock must be flagged. Delayed or unavailable data must be disclosed. Stale prices must never be shown as current.

4.3. Financial News

The system must collect news and announcements on earnings, revenue and profit, mergers, management changes, regulation, governance issues, industry news, government policy, interest rates, inflation, currency and commodity moves, and global events affecting Indian stocks. Official company and exchange announcements get priority.

For every important news item, store the **source, link, publication date and time, affected stock or sector, sentiment, and possible impact**. Confirmed information must be separated from rumors and unverified reports. News older than a set number of days (configurable) must not be used for the five-day outlook.

5. Daily Analysis

5.1. Price and Volume Analysis

The system must study short-term trends, momentum changes, unusual trading activity, breakouts and breakdowns, volatility, support and resistance levels, reversals, and whether volume confirms price moves. It must not assume history will repeat.

5.2. Technical Analysis

The system must calculate SMA, EMA, RSI, MACD, Bollinger Bands, ATR, support and resistance, momentum, and volume trends. It must combine several signals and never decide from one indicator alone. It must identify bullish and bearish signals, overbought conditions, weakening momentum, and excessive volatility.

5.3. Fundamental Analysis

The system must check revenue growth, profit growth, EPS, P/E, P/B, ROE, ROCE, debt-to-equity, margins, cash flow, financial stability, and valuation against industry peers. It must consider sector differences and must not call a stock fundamentally strong only because its price recently rose.

5.4. Sentiment Analysis

Sentiment is classified as Positive, Negative, Neutral, or Mixed/Uncertain. The system must notice changes in sentiment and explain serious negative news even when technical signs look positive.

5.5. Overall Market Conditions

Before recommending any stock, the system must study Nifty 50 and Nifty 100 performance, sector moves, global markets, domestic economy, central bank announcements, interest rates, inflation, and geopolitical events. If the overall market is weak, it must not blindly recommend stocks that only look bullish.

6. Stock Screening and Five-Day Prediction

6.1. Screening and Ranking

The system ranks stocks using trends, technical indicators, volume, fundamentals, news, sentiment, sector performance, valuation, volatility, downside risk, and liquidity. The developer must define **configurable thresholds and a transparent scoring method**. Stocks must not be ranked only by predicted percentage gain.

Extra rules: a **minimum liquidity filter** so hard-to-trade stocks are not suggested, and a **maximum number of recommendations per day** (configurable).

6.2. Prediction Method

The developer must clearly define how the forecast is made, whether by a statistical model, a machine learning model, or an AI model reasoning over calculated numbers. The expected price range should be based on measurable data such as recent volatility (for example ATR or standard deviation of returns). Prices must not be guessed directly by an AI model without a measurable basis. The model version must be stored with every forecast.

6.3. Required Output for Each Recommended Stock

Stock name and symbol, exchange, reference price and its timestamp, predicted direction, expected price range after five trading days, expected percentage change, technical summary, fundamental summary, news and sentiment summary, supporting reasons, downside risks, confidence or uncertainty, and support and resistance levels.

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\text{Expected Appreciation (\%)} = \frac{\text{Expected Price} -
\text{Reference Price}}{\text{Reference Price}} \times 100
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For the pre-market report, the reference price is the previous completed trading session's closing price. The forecast is an estimate, never a guaranteed target.

6.4. Confidence in the Early Period

Confidence must be based on validated model performance. In the first weeks there is not enough history, so the report must show "Not enough history yet to measure accuracy" instead of a made-up confidence number.

6.5. Risk Level Rule

Each stock gets a Low, Medium, or High risk level based on written rules (for example volatility and past drawdown), so the label is not a matter of opinion. The developer defines the rules and shows them in the report's methodology section.

6.6. Recommendation Rules

The system recommends stocks only when the screening criteria are met. It considers downside risk before ranking, avoids recommendations when essential data is missing, explains the main reasons, points out conflicting signals, and clearly reports when **no strong opportunity** is found. It must never create recommendations just to fill the report.

7. Tracking and Evaluation of Recommendations

7.1. Original Recommendation Records

For every recommendation, store a unique ID, stock and exchange, date, original reference price, original expected range and percentage, horizon, risk level, supporting technical and fundamental information, relevant news, methodology and model version, and creation timestamp. These records are never overwritten, even if the stock is later downgraded or removed.

7.2. Daily Monitoring

Every day the system compares earlier recommendations with the latest data. It finds stocks that fell, lost expected upside, show bearish signals, face negative news, show weaker fundamentals, carry higher downside risk, still look favorable, or no longer meet the criteria. A small price drop alone must not cause a downgrade. The decision must consider price, technical signals, news, fundamentals, market conditions, and the original forecast.

7.3. Statuses

Status	Meaning
Positive — Outlook Maintained	Stock still meets the criteria
Watchlist — Potential Weakness	Early warning signs, reduced upside, or mixed signals

Status	Meaning
Downgraded — Negative Outlook	The positive outlook has clearly weakened
Removed — No Longer Meets Criteria	Stock no longer passes the screening
Insufficient Data	Not enough data for a reliable reassessment

Every important status change must come with a written reason.

7.4. Exact Five-Day Rule

The system must follow this rule, which the developer may refine only with the owner's approval. A report is created on the morning of trading day T using the close of the previous session. **Day 1 is the first trading session on or after the report date. The evaluation point is the closing price of the fifth trading session.** Weekends and exchange holidays are skipped using the official trading calendar. Not every recommendation ends on a Friday.

If a stock is halted or hits a circuit limit on the evaluation day, the evaluation must be marked as an exception and handled by a written rule.

7.5. Forecast Evaluation

After the five-trading-day period, the system keeps the original reference price and forecast, records daily observations, finds the endpoint, gets the actual price, and calculates actual return, forecast error, and whether the direction was right. Results are stored permanently. Revised forecasts must never replace the original when measuring accuracy.

The system must measure directional accuracy, price error, actual versus predicted return, results in different market conditions, the frequency and size of downside surprises, and results over different periods.

Baseline comparison: every evaluation must also show the **Nifty 50 index return over the same five days**, so the owner can see whether the recommendations beat simply holding the index.

Reference price versus real buy price: because the user usually buys after the market opens, the real buy price can differ from the previous close. For held stocks, the evaluation must show results from both the original reference price and the user's actual purchase price.

7.6. Backtesting Before Going Live

Before real use, the developer must test the scoring and prediction method on at least the past two to three years of data and report the results against the Nifty 50 baseline. The system must not go live until the owner has seen these results.

The system must not claim that every recommendation made a profit.

8. Execution Schedule

8.1. Three Layers of Work

Layer	When	Job
Light Monitor	Continuously, every few minutes (not every second)	Check new news, announcements, and large price moves for recommended and held stocks.
Main Analysis	Once after market close (market closes 3:30 PM IST, so start after final data is available, around 4:30–5:00 PM)	Calculate indicators, score stocks, create forecasts, update tracking and portfolio values.
Morning Refresh	Early morning, around 4:30 AM IST	Add overnight news and global market moves, and update forecasts if something important changed.

Running the full analysis every second is not needed. Daily price data changes only once a day, free data sources have request limits, and many free prices are delayed. Constant heavy analysis would waste resources and may cause blocks or errors.

8.2. Report Timeline (target milestones, not guarantees)

Time (IST)	Step
About 5:15 AM	Start building the HTML report
About 5:25 AM	Create the PDF
About 5:35 AM	Validate that the report is complete
5:45 AM	**Send the email (deadline)**

If a required part has failed by 5:35 AM, the system still sends the report on time and clearly states which part is incomplete. It must never send a report that falsely says all checks passed. The email is sent only after the report is generated and validated.

8.3. Trading Days Only

The report is sent only on mornings when NSE is open that day. The system uses the **official NSE trading calendar**, updated every year, not just "Monday to Friday," because special sessions can occur on weekends or holidays. On weekends and holidays no report is sent unless the owner changes this setting.

8.4. Pre-Market Data

At 5:45 AM the exchanges have not opened. The system uses the previous session's closing prices, overnight news, available international market information, economic announcements, and other reliable pre-market data. The report must show the latest data timestamp.

9. Portfolio Dashboard

9.1. Purpose

A separate Portfolio Dashboard is mandatory. The user records real investments there, not through the emailed PDF, and never has to re-enter the same holdings. The dashboard must stay available at all times, independent of the scheduled workflow.

9.2. Dashboard Sections

Today's Stock Recommendations, Investment Input Form, My Portfolio, Portfolio Performance, Portfolio Risk Alerts, Investment Transaction History, and Historical Recommendation Tracking. The design must be clear, responsive, and easy to use.

9.3. Today's Recommendations

For each stock show the name, symbol and exchange, reference price, expected price after five trading days, expected appreciation, risk level, and status. Each stock has a selection option so the user can say whether they invested. The system must never assume the user bought a stock only because it was recommended. The user may invest in none, some, or all.

9.4. Investment Input Form

Field	Description
Stock	Selected stock
Exchange	NSE or BSE
Investment Date	Date of purchase
Purchase Price	Actual price paid per share
Quantity	Number of shares bought
Investment Amount	Total purchase value
Transaction Costs	Optional brokerage and other costs
Notes	Optional comments

The system calculates total value from quantity and price and checks the entered amount. If the user enters only an amount, the system may estimate the quantity but must clearly label it as **estimated**, and ask the user to confirm the real price and quantity.

9.5. Investment Management

The user can add investments, correct details, record extra purchases, record partial or full sales, review past transactions, view current holdings, see the original recommendation linked to an investment, and see the latest assessment. Corrections

must not silently destroy history. Extra purchases and sales are stored as separate transactions. Multiple purchases of the same stock at different prices must be supported. The system keeps full transaction history, not just the latest balance.

9.6. Persistent Storage

Store the portfolio ID, investment ID, stock and exchange, related recommendation ID, date, price, quantity, amount, costs, sales, remaining holdings, and timestamps. Saved holdings load automatically when the user returns and survive restarts and new reports.

9.7. Example

Monday's report recommends five stocks and the user buys four:

Stock	Purchase Price	Quantity	Amount
Stock A	₹1,000	10	₹10,000
Stock B	₹500	10	₹5,000
Stock C	₹800	10	₹8,000
Stock D	₹200	20	₹4,000
Stock E	—	—	Not purchased

Total recorded investment: ₹27,000.

10. Portfolio Monitoring and Indian Market Rules

10.1. Metrics for Each Holding

Total invested amount, quantity held, average purchase price, latest price, current market value, unrealized profit or loss (amount and percentage), realized profit or loss from sales, original expected price, updated expected range where justified, current status, risk level, and important positive or negative developments.

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\text{Unrealized P\&L} = \text{Current Market Value} - \text{Remaining Cost Basis}
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Realized and unrealized profit must be calculated separately.

10.2. Cost Basis and Costs

The system uses **FIFO** for stocks with several purchases and partial sales. It must show profit **before and after costs**, including brokerage, STT, exchange charges, GST, and stamp duty. It should also indicate short-term and long-term capital gains status of sold shares so the user can understand possible tax. Final tax calculation remains the user's responsibility, and the report must say so.

10.3. Corporate Actions

The system must adjust quantity, cost basis, and stored price history for **stock splits and bonus issues**, and record **dividends** as income. Without this, profit and loss numbers would be wrong.

10.4. Example: Tuesday Portfolio

Stock	Purchase Price	Qty	Latest Price	Market Value	Unrealized P&L
Stock A	₹1,000	10	₹1,020	₹10,200	+₹200
Stock B	₹500	10	₹460	₹4,600	−₹400
Stock C	₹800	10	₹790	₹7,900	−₹100
Stock D	₹200	20	₹215	₹4,300	+₹300
Total				**₹27,000**	**₹0**

These are example numbers without costs or corporate actions. The report must explain that Stock B fell and needs more analysis while Stock D rose. A gain does not automatically mean a stock is still a good opportunity, and a loss does not automatically mean it should be sold.

10.5. Risk Alerts

The system must explain the evidence behind every portfolio alert. It must also give a **sector concentration warning** if too much of the portfolio is in one sector.

11. Recommendations and Holdings Stay Separate

Every report has two separate views.

Today's Stock Recommendations shows all stocks that currently meet the criteria, whether or not the user owns them. Each is marked as already held, newly recommended and not held, or previously recommended but no longer shortlisted. A stock appearing again on a later day is not automatically a new purchase.

My Investment Portfolio shows only the user's recorded holdings, with purchase price, quantity, invested amount, latest price, market value, realized and unrealized profit or loss, current assessment, updated forecast where justified, and risks. It never includes stocks the user has not bought, and a holding never disappears because the agent removed it from the shortlist.

12. Daily Workflow (Monday to Friday Example)

Monday: The agent analyzes the market, creates recommendations and five-day forecasts, builds the report, and emails it by 5:45 AM. The user opens the dashboard, selects the stocks bought, and enters real prices and quantities. Forecasts and investment records are stored separately.

Tuesday: The agent collects new data, creates fresh recommendations, retrieves Monday's recommendations, checks changes, loads the saved portfolio, calculates values and profit or loss, finds possible downgrades, updates assessments without touching original forecasts, and emails the new report. The user does not re-enter anything.

Wednesday: Fresh recommendations, tracking of earlier recommendations, portfolio update, alerts for weakening holdings, stocks with improved outlook, and tracking of forecasts not yet at their endpoint.

Thursday: The same process, with special focus on holdings hit by negative news, worsening indicators, or higher downside risk. Holdings that left the shortlist still appear.

Friday: A new report, portfolio tracking, and evaluation of forecasts whose five-trading-day periods ended. The report separates new recommendations, existing holdings, weakening earlier recommendations, pending forecasts, and completed forecasts with actual outcomes. The system follows the trading calendar and does not assume every forecast ends on Friday.

13. Report Content

The HTML and PDF versions must contain the same substantive information.

Section	Contents
1. Executive Summary	Report date and generation time, latest data timestamp, market outlook, key developments, number of stocks analyzed, number of new recommendations, major portfolio changes, downgrade alerts, significant risks
2. Market Overview	Nifty 50 and Nifty 100 performance, sector trends, global markets, economic news, sentiment, market risk
3. Today's Top Recommendations	Name, symbol, exchange, reference price, expected range and appreciation, risk level, confidence or uncertainty
4. My Investment Portfolio	Holdings, prices, quantities, invested amount, market value, realized and unrealized P&L (before and after costs), portfolio performance, risk alerts
5. Previous Recommendations — Performance and Downgrade Alerts	Original price, latest price, change since recommendation, change since previous session, original range, updated forecast if justified, current status, reason for change, negative developments, risks
6. Updated Five-Day Forecasts	Latest forecasts for earlier recommendations, with material changes from the originals
7. Completed Forecast Evaluation	Original predicted price, actual price, predicted and actual return, error, direction result, Nifty 50 baseline return. Unfinished forecasts are marked pending
8. Market Risks and Upcoming Events	Economic releases, earnings dates, corporate events
9. Data Sources and Methodology	Sources, timestamps, prediction method, screening rules, risk rules, limitations

Section	Contents
10. Disclaimer	Analysis uses available information, forecasts are uncertain, returns are not guaranteed, not a guarantee of future performance

Design: clear headings, readable tables with borders, consistent fonts, green for positive moves, red for negative moves, neutral style for mixed signals, price charts where useful, print-friendly layout, proper PDF page breaks, and generation timestamps. The HTML may use interactive charts.

14. Email Delivery

Subject: Daily AI Stock Market Analysis and Portfolio Report — [Report Date]

Delivery: by 5:45 AM IST on market days.

Attachments: the PDF report and, where suitable, the HTML report.

Email body: a short summary of the market outlook, top opportunities, current portfolio value and performance, major negative developments, downgraded stocks, important forecast changes, and significant risk alerts, directing the reader to the attachment for details.

The system records report generation and delivery status, retries recoverable failures, logs unresolved failures, and avoids duplicate emails for the same scheduled run.

15. Data Storage and Architecture

15.1. Storage

Persistent storage (PostgreSQL is a suggestion, the developer decides) holds these logical groups: **Stock Recommendations** (original forecasts), **Daily Recommendation Updates** (daily monitoring results), **Portfolio Transactions** (purchases and sales), **Portfolio Daily Snapshots** (daily valuations with data timestamps), and **Forecast Evaluations** (completed outcomes). Records are linked with stable IDs. Original recommendations are never overwritten, transactions are not changed when a status changes, and historical snapshots keep the data used for old reports so past results can be reproduced and audited.

15.2. Backups

The database must have **automatic daily backups** and a tested way to restore them, because losing original forecasts or transactions would break the core purpose of the system.

15.3. Agent Structure

The developer may use these modules: Market Data, Financial News, Sentiment Analysis, Technical Analysis, Fundamental Analysis, Stock Screening, Prediction, Risk Assessment, Historical Tracking, Portfolio Management (uses only confirmed user transactions), Forecast Evaluation, Report Generation, Email Notification, and an Orchestrator that coordinates the sequence, validates outputs, and handles recoverable failures. The dashboard and the automated workflow share the same stored records.

16. Error Handling, Security, and Monitoring

Data and errors: validate all market data, detect missing, delayed, or inconsistent information, avoid forecasts when critical data is missing, record execution status and errors, retry recoverable failures, and detect duplicate runs. If old recommendation records cannot be read, the report must say historical comparison could not be completed. If portfolio data is unavailable, the system must not invent holdings or calculate results from assumed investments.

Security: the dashboard requires login with password plus two-step verification, all connections use HTTPS, and API keys and email credentials are stored securely and never exposed. Portfolio information must never appear in public outputs. The developer must confirm whether the system serves one user or many, since this changes the database design.

System monitoring: a separate short alert (email or message) must be sent to the owner if the daily job does not start, or if the report email has not been sent by a set deadline shortly after 5:45 AM. Without this, a silent failure could go unnoticed.

17. Non-Functional Requirements

Quality	Meaning
Reliability	Daily execution, persistent storage, retries, error monitoring
Data Quality	Reliable data with clear timestamps and sources
Transparency	Every recommendation has evidence and assumptions
Security	Protect portfolio data, API keys, and email access
Scalability	Easy to add more stocks and markets
Maintainability	Data sources and analysis parts can be changed independently
Auditability	Original forecasts, transactions, and report records are preserved
Consistency	HTML and PDF show the same substantive information
Performance	Analysis and delivery finish within the time window
Usability	Simple dashboard for entering and reviewing investments

18. End-to-End Workflow

1. The light monitor runs continuously and the main analysis runs after market close.
2. The orchestrator starts the analysis run.
3. Market data and news are collected and checked against the backup source.
4. Technical, fundamental, and sentiment analysis run on the stock universe.
5. Stocks are screened, ranked, and forecasted.
6. Earlier recommendations are retrieved and checked for material changes.
7. Confirmed holdings and transactions are loaded and valued.
8. Stock and portfolio risks are assessed.
9. Completed forecasts are evaluated against actual outcomes and the Nifty 50 baseline.

10. The morning refresh adds overnight news and global moves.
11. The orchestrator validates results and notes anything missing.
12. The HTML report is built, then converted to PDF.
13. The email is sent by 5:45 AM on market days.
14. Execution status, data timestamps, report references, and delivery results are recorded.

After any new transaction is saved on the dashboard, all later reports use the updated portfolio.

19. Expected Final Outcome

The user can receive fresh Indian stock recommendations every market morning, review predicted five-day price ranges, see technical, fundamental, and sentiment analysis, track earlier recommendations and downgrades, keep original predictions for honest evaluation, record real purchases and sales on the dashboard, monitor portfolio value and profit or loss, receive alerts when holdings weaken, compare completed predictions with real outcomes and with the Nifty 50, and receive a detailed HTML and PDF report by email.

The system must clearly separate AI recommendations, user-confirmed investments, current assessments, and completed forecast evaluations.

Final Objective: Build a reliable, automated system that delivers daily, evidence-based stock recommendations before the Indian market opens, tracks the user's real investments, finds changing conditions and downside risks, evaluates past predictions honestly, and gives a continuously updated view of opportunities and portfolio performance. It must prioritize transparent analysis, accurate records, reliable data, and risk awareness. It must never guarantee returns, invent missing information, or overwrite historical forecasts to make its performance look better.